

Resolving symmetry without constraining alternatives *

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Abstract

In cases where an assertion could in principle lead to several, mutually incompatible quantity implicatures, some of these implicatures seem systemically unavailable. Katzir (2007) and Fox and Katzir (2011) aim to solve this so-called ‘symmetry problem’ by preventing implicatures that are based on the exclusion of alternatives that are syntactically more complex than the assertion itself. We argue that this complexity filter on alternatives falls short of solving the symmetry problem completely, and in fact is incompatible with the full range of observed implicatures. We propose a solution to the symmetry problem that, in addition to contextual relevance, appeals to a blocking condition: enriching the meaning of an utterance ϕ with an implicature is blocked if the same meaning can be expressed without this implicature, by an utterance that is no more complex than ϕ . In making our case, we also argue against a central auxiliary assumption that Katzir and Fox’s account appeals to, viz. the assumption that symmetry cannot be resolved by context.

Keywords: exhaustification, quantity implicature, symmetry problem, complexity constraint, blocking

1 Introduction

Exhaustification is a process that can tacitly strengthen basic sentence meanings. Prominent instances of this effect are so-called scalar implicatures, inferences routinely associated with uses of certain logical vocabulary. Sentences with the existential determiner *some* are a familiar case in point. Sentence (1) has a basic meaning that is merely existential, that there are players who responded, which we will state as in (2). But (1) can, and typically does, carry the stronger total meaning in (3). That is, (1) can be understood as also implying that there were players who did not respond, the second conjunct of (3). The possibility of this implication is confirmed, for instance, by its felicitous denial, as in (4).

(1) Some players responded.

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- (2) Basic meaning of (1):
SOME(P)(R)
- (3) Available strengthened meaning of (1):
SOME(P)(R) $\wedge$ $\neg$ ALL(P)(R)
- (4) A: Some players responded.
B: No, all of them did!

The nature of exhaustification is a matter of debate. In particular, while the mapping from basic to strengthened meaning is classically attributed to Gricean pragmatics, it has also been credited to a grammatically encoded operator (e.g., Chierchia et al. 2011). However, since the choice between these options does not bear on the arguments in this paper, we will refrain here from committing to one view or another. Reflecting this lack of commitment, we will often use the terms *strengthening* and *strengthened meaning* to refer to the phenomenon in question, which might be more obviously theory neutral than *exhaustification*. At the same time, it will be convenient to also have a name for the inferences that strengthening adds to basic meaning, such as the second conjunct in (3), and we will employ the term *implicature* to this effect.

In the broadest characterization, strengthening takes the form in (5). That is, the strengthened meaning of sentence ϕ , $\text{Exh}_{\text{Excl}}(\phi)$, is the conjunction of the basic meaning of ϕ , $\text{Basic}(\phi)$, with the negations of the members of a set of propositions *Excl*. We will also refer to *Excl* as the *exclusion set*, and to its members as the *excluded propositions*. The format in (5) can be illustrated with the equality in (6). That is, the strengthened meaning of sentence (1) given in (3) can be understood as due to strengthening relative to an exclusion set that contains just the proposition ALL(P)(R), that all players responded.

- (5) Strengthened meaning:
 $\text{Exh}_{\text{Excl}}(\phi) = \text{Basic}(\phi) \wedge \bigwedge \{ \neg p : p \in \text{Excl} \}$
- (6) $\text{Exh}_{\{ \text{ALL(P)(R)} \}}(1) = \text{SOME(P)(R)} \wedge \neg \text{ALL(P)(R)}$

A central task for a theory of strengthening is to explain limitations on the range of actually observed sentence meanings of the format in (5). To illustrate one important type of limitation, we can continue to focus on sentence (1). Consider a hypothetical exclusion set that only contains $\neg \text{ALL(P)(R)}$, the proposition that not all players responded. As stated in (7), strengthening (1) with this exclusion set would yield ALL(P)(R), the proposition that all players responded. As recorded in (8), however, such a strengthening is unavailable for (1). This is confirmed by the observation in (9) that the relevant hypothetical implicature cannot be felicitously denied.

- (7) $\text{Exh}_{\{ \neg \text{ALL(P)(R)} \}}(1) = \text{ALL(P)(R)}$
- (8) Unavailable strengthened meaning of (1):
ALL(P)(R)
- (9) A: Some players responded.
B: #No, not all of them did!

Why is it that the basic meaning of (1) can be strengthened with the implicature $\neg \text{ALL(P)(R)}$, yielding (3), but not with ALL(P)(R), yielding (8)?

Sometimes context can have the effect that an implicature that is in principle possible is not in fact attested. For example, in a context in which it is not relevant whether *some but not all* or *all* players responded, and all that matters is simple whether *some* did or not, the implicature in (3) may not materialize. But we follow previous literature in the conclusion that relevance will not be able to explain why (3) is available but (1) is not. The inference $\neg\text{ALL}(\text{P})(\text{R})$ in (3) presumably arises only when this inference is relevant, and when the excluded proposition $\text{ALL}(\text{P})(\text{R})$ is relevant. But if both of these meanings are relevant, then it remains unclear why the inference in (8) is not viable (cf. Fox 2007).

Why is it, then, that some conceivable implicatures are impossible, even though their symmetric counterparts are attested? For any implicature that we observe to be possible, we can in fact easily create a symmetric competitor, simply by adding negation:

(10) Conceivable symmetric strengthenings:

- a. $\text{Basic}(\phi) \wedge \neg p$
- b. $\text{Basic}(\phi) \wedge p$

Whenever the basic meaning of a sentence ϕ can be strengthened with the conjunct $\neg p$, as in (10a), but not with the conjunct p , as in (10b), the question arises on what grounds the former is available while the latter is not. Since the two strengthenings in (10) and the implicatures they add are in a transparent sense *symmetric*, this problem is known as the *symmetry problem*. First identified by Kroch (1972) with reference to a Gricean view of strengthening, the symmetry problem has figured prominently in recent work on exhaustification.

1.1 Filtering alternatives

The dominant approach to address this issue involves imposing constraints on the membership of the exclusion set Excl . On this view, if for a given sentence ϕ , the strengthening (10a) is attested while (10b) is not, this is assumed to be due to a constraint that allows for an exclusion set that yields (10a), but prohibits an exclusion set that would yield (10b). For example, sentence (1) is assumed to permit the strengthening in (3), but not the one in (8), because the exclusion set for the interpretation of (1) may contain $\text{ALL}(\text{P})(\text{R})$ to yield (3), but it cannot contain $\neg\text{ALL}(\text{P})(\text{R})$, or any other proposition that would enable the derivation of (8).

This approach has prominently been developed by constraining the exclusion set with reference to the *linguistic form* that would be required to express certain meanings. The idea is that for strengthening of ϕ , any proposition in the exclusion set must be the basic meaning of a sentence, a *linguistic alternative* to ϕ , that is no more complex than ϕ itself. Horn (1972), for example, postulates lexical scales, today often called *Horn scales*; Levinson (1983) invokes an *economy* condition; and Horn (2000) a *brevity* condition. More recently, Katzir 2007 and Fox and Katzir 2011 have formalized a notion of complexity that requires formal alternatives to be no more complex than the asserted sentence. This *complexity filter on Excl* (CF) can be stated as in (11).

(11) Complexity filter on Excl (CF):

$$\text{Excl}(\phi) \subseteq \{\text{Basic}(\psi) : \psi \text{ is no more complex than } \phi\}$$

The complexity filter has the intended effect for sentence (1). As noted, the unavailable strengthened meaning (8) could potentially arise for (1) from excluding the proposition $\neg\text{ALL}(\text{P})(\text{R})$. This

proposition is the basic meaning of, for example, the sentence in (12). However, the presence of negation renders (12) more complex than (1). In fact, it appears that any linguistic encoding of $\neg\text{ALL}(\text{P})(\text{R})$, or any other meaning that would yield the strengthening in (8), will require added complexity relative to (1). If so, then CF correctly prevents this strengthening.

(12) Not all players responded.

In fact, CF generally diffuses the recipe for creating a problematic symmetric alternative by adding negation: Since negation adds complexity, the competing alternative created by this recipe should never be viable.

In this paper, we will build a case against this approach to the symmetry problem. On the one hand, we will argue that CF is too demanding a condition on Excl. In some cases, the sorts of linguistic alternatives that CF prohibits would actually be needed to explain strengthenings that are intuitively available. On the other hand, extending observations in the previous literature, we will argue that CF is also insufficient, sometimes permitting alternatives that yield strengthenings which are not in fact attested.

1.2 Blocking strengthened outputs

If we are correct in arguing that imposing a syntactic complexity filter does not yield a solution to the symmetry problem, then what does? We propose that using a sentence with a strengthened reading is blocked if there is an equally or less complex sentence that conveys the strengthened meaning without implicature. To illustrate the idea, consider once again sentence (1), and let us juxtapose it with (13).

(13) All players responded.

Note that the basic meaning of (13) is identical to the unavailable strengthened meaning for (1) in (8). What we propose is that, since (13) is moreover no more complex than (1), its existence *blocks* the use of (1) in the meaning in (8), rendering this strengthening unavailable. So we propose that a potential strengthened meaning for a sentence ϕ is unavailable if it amounts to the *basic* meaning of a sentence no more complex than ϕ . We can state this *blocking constraint* (BC) as in (14).¹

(14) Blocking constraint (BC):

$\text{Exh}_{\text{Excl}}(\phi)$ is blocked if: $\exists \psi [\text{Basic}(\psi) = \text{Exh}_{\text{Excl}}(\phi) \wedge \psi \text{ is no more complex than } \phi]$

BC effectively demands that strengthening always be in the service of abbreviation—a sort of economy constraint that could perhaps be grounded in Grice’s Manner maxim. That is, BC demands that strengthening serve to *reduce complexity* of linguistic form that would otherwise be required to express a given total meaning. Given that (1) is no less complex than (13) is, the unavailable strengthening for (1) in (8) is blocked by (13) on those grounds.

In contrast, even though the proposition in (3) is the basic meaning of sentence (15), (1) is less complex than (15) is.

(15) Some but not all players responded.

¹In Appendix A, we discuss cases that reveal that the blocking condition in (14) is not general enough, and we propose a refinement. This revision does not affect the arguments presented in the main text.

It in fact seems that *any* sentence of English with this basic meaning is more complex than (1). If so, then, the available strengthening in (3) enables abbreviation, and is correctly permitted by BC.

Before we turn to arguments that distinguish CF and BC, we should note that the idea that blocking plays a role in exhaustification is not new. Blocking was invoked, for example, in the account of quantity implicatures presented in Geurts (2011: ch. 6). Geurts appeals to complexity to a account for a listener’s selection of cells from a partition of the speaker’s “intentional state”. *Some players responded* cannot convey ALL(P)(R), because to convey the corresponding intentional state, it would have been more expedient for the speaker to assert *All players responded*. This approach is similar to ours in appealing to the complexity of alternative assertions in reasoning about the likely intended meanings. It differs in that the account does away to any appeal to the exclusion of alternative meanings, and in that sense departs even more radically from accounts that try to constrain the set of excluded alternatives.

Another precedent for applying the logic of blocking to scalar implicatures is implicit in the Rational Speech Act (RSA) framework (see Goodman and Frank 2016, Degen 2023 for overviews). The RSA models a speaker’s choice for how to convey a particular meaning as an optimization problem. Two factors are central for each decision on how to convey a meaning: One is the likelihood that an utterance will convey the correct meaning successfully; the other is the production cost of a given utterance, which increases with syntactic complexity. So if the intended meaning is ALL(P)(R), then using *Some players responded* instead of *All players responded* would preserve complexity and hence not serve to lower production cost, and at the same time would be less likely to successfully communicate ALL(P)(R) than *All players responded*, as it could just convey SOME(P)(R). *Some players responded* will therefore never be the speaker’s preferred way of conveying ALL(P)(R).²

Both of the alternative accounts referenced above incorporate blocking reasoning parallel to the one we proposed in our formalization of BC. We will not attempt to compare them accounts to ours in this paper, and it is quite possible that the arguments below in favor of BC are in fact arguments for a broader class of accounts that appeal to blocking in explaining constraints on quantity implicatures.

1.3 Differentiating filtering and blocking

When it comes to explaining why *some* cannot mean ALL, we have seen that either CF or BC could have the intended effect. But how can these accounts be teased apart empirically? This is not obvious, in particular since the notion of complexity that BC appeals to may well be the same notion as the one invoked by CF, and the two conditions seem closely related.

In this paper, we will discuss two ways in which the accounts can be differentiated. Katzir (2007) and Fox and Katzir (2011) propose CF as a general constraint on Excl, not as a constraint that only applies to alternatives with symmetric counterparts. Blocking, by contrast, can only arise in cases of symmetry. In Section 2, we argue that CF cannot be maintained as a general constraint on

²Another precedent can be found in Chierchia (2013: 129), who argues for ‘Exhaustification Economy,’ which constrains Exh so that it cannot derive meanings that are equivalent to an already existing alternative in the set of alternative meanings. Chierchia uses this constraint explain why disjunctions cannot be strengthened to convey *and*, but does not discuss further implications for other cases of symmetry, or the symmetry problem in general.

Excl, because complex alternatives are in fact often needed to derive attested implicatures. We also show that BC is compatible with these cases, and in fact makes interesting novel predictions about them. This means that CF can at best be a tie-breaker constraint for symmetric alternatives, but not a general formal constraint on the alternative in Excl, let alone alternative substitution more generally.

The two accounts diverge in another way. In Section 3, we will look at cases in which an unavailable implicature would be derivable based on alternatives that are not more complex than the asserted sentence. These cases show that CF is insufficient to explain the symmetry problem. We will also show that BC provides a straightforward account for these cases.

To further investigate the viability of CF, Section 4 scrutinizes the effects of certain auxiliary assumptions employed in Katzir (2007) and Fox and Katzir (2011) to address apparent counterexamples. A central role in making CF viable is played by a further condition on Excl that we will call *stalemate assumption*. Suppose that a pair of symmetric alternatives are both compatible CF and could hence be part of Excl. The stalemate assumption holds that in this case, it is impossible, regardless of context, to selectively omit one of them in order to derive an implicature from the other. According to Katzir (2007) and Fox and Katzir (2011), symmetry effectively creates a stalemate between two possible but mutually incompatible strengthenings, unless CF can break the tie. We will argue that, despite the interesting empirical motivation for this condition, the stalemate assumption cannot be maintained, and in particular that stalemates can be broken by blocking.

We will also show that in cases where neither blocking nor context eliminates relevant alternatives, empirical ‘stalemates’ can arise, but argue that such cases call for a pragmatic explanation, not a formal condition such as the stalemate assumption. Section 5, finally, looks at some apparent counterexamples to BC, and argues that they may in fact provide further support for BC and against CF and the stalemate assumption. Appendix A shows that the blocking condition as stated in (14) is not general enough, as it undergenerates blocking, and we propose a generalized version of BC that has the intended effect. Appendix B explores the logical relation between CF and BC in their effects on symmetric alternatives.

2 The case against the complexity filter

The symmetry problem amounts to the question why a given sentence permits just one of two conceivable mutually incompatible strengthened meanings. We saw that the solution based on the complexity filter CF constrains the exclusion set Excl, by preventing certain linguistic alternatives from contributing to its membership. Let us review, for ease of reference, our illustration of this from the introductory section. We saw that for the exhaustification of (1), repeated below as (16), CF allows for Excl to contain the basic meaning of (17a), and thereby permits the strengthened meaning in (18a). In contrast, CF prevents Excl from including a proposition whose exclusion would yield (18b), on the grounds that any sentence that has such a proposition as its basic meaning, including (17b), is more complex than (16).

(16) Some players responded.

(17) a. All players responded.

- b. Not all players responded.
- (18)
- a. $\text{SOME}(\text{P})(\text{R}) \wedge \neg \text{ALL}(\text{P})(\text{R})$
 - b. $\text{ALL}(\text{P})(\text{R})$

An attractive feature of the account is that the condition on Excl imposed by CF is *general* in the sense that it does not in any way reference the notion of symmetry. The expected effects of CF therefore are not just about pairs of mutually incompatible potential strengthened meanings. One expects to find cases where CF permits only one of two conceivable implicatures even though the two are compatible with one another, and hence could in principle jointly contribute to a contingent and observable strengthened meaning. This line of thought suggests an avenue for scrutinizing CF which we will now pursue.

Adapting an example that Fox and Katzir (2011) present for a somewhat different purpose, consider sentence (19) below, as well as its conceivable linguistic alternatives given in (20a,b). We will show that CF makes predictions about (19) and (20) that parallel the predictions for (16) and (17), notwithstanding the fact that the two cases have different logical profiles.

- (19) The detective concluded that Jed stole some of the books.
- (20)
- a. The detective concluded that Jed stole all of the books.
 - b. The detective concluded that Jed did not steal all of the books.

The embedded clauses of (20a,b) relate to the embedded clause of (19) in the same way that the alternatives in (17a,b) relate to (16). That is, both (20a) and (20b) differ from (19) in that *all* has been substituted for *some* in the embedded clause, but (20b) in addition departs from (19) in that the embedded clause is negated. Accordingly, while sentence (20a) is just as complex as (19), the presence of negation renders (20b) more complex than (19). Basic(20b) moreover does not seem to permit a linguistic encoding that avoids added complexity relative to (19). If so, then, for the exhaustification of (19), CF allows for Excl to contain Basic(20a) to yield the implicature $\neg \text{Basic}(20a)$, but prevents Excl from containing Basic(20b) to derive $\neg \text{Basic}(20b)$. According to CF, then, in parallel to the predictions for (16), it should be possible to understand sentence (19) as denying that the detective concluded that Jed stole all of the books, but not as denying that the detective concluded that Jed did *not* steal all of the books.

Despite this parallel between the cases of (16) and (19), though, they differ in their logical profile in the way previewed above. While the potential implicatures $\neg \text{Basic}(17a)$ and $\neg \text{Basic}(17b)$ of sentence (16) are each other's negations, hence cannot be true at the same time, the potential implicatures $\neg \text{Basic}(20a)$ and $\neg \text{Basic}(20b)$ of sentence (19) are mutually compatible: it is logically possible for the detective to be agnostic about whether Jed stole all of the books, concluding neither that he did nor that he didn't. This compatibility is made possible by the embedding predicate *conclude*, which for any given proposition, can consistently relate an attitude holder to neither that proposition nor its negation. In this sense, *conclude* removes the symmetry that may hold between two given complement clauses.

Yet again, CF is blind to this difference in logical profile between cases of (16) and (19), as it applies to the two cases in a parallel way. But is this actually correct? To assess the relevant intuitions, (21) presents (19) as a response to a sentence that should render the basic meanings of the alternatives in (20) relevant. We intend this to establish a precondition for the inclusion of those

meanings in Excl, given the consensus that Excl only includes relevant propositions. Now, what is sentence (21B) intuited to convey?

- (21) A: Tell me what exactly the detective has concluded up to this point.
B: The detective concluded that Jed stole some of the books.

We submit that (21B) can be understood to carry either of the two implicatures $\neg$ Basic(20a) and $\neg$ Basic(20b), and in fact could convey their conjunction—that the detective is agnostic about whether Jed stole all of the books. This verdict is supported by the intuition that it is intuitively felicitous to deny either implicature in conversation. The replies in (22a) and (22b), which we can imagine being uttered by a third person, diagnose $\neg$ Basic(20a) and $\neg$ Basic(20b), respectively.

- (22) a. No, the detective also concluded that Jed stole all of the books!
b. No, the detective also concluded that Jed did not steal all of the books!

So it seems that, while CF is correct in allowing for (19) to carry the implicature $\neg$ Basic(20a), its prediction that (19) cannot implicate $\neg$ Basic(20b) is incorrect. This suggests that by preventing Excl from containing Basic(20b), CF is too restrictive a condition on Excl.³

Observations supporting this verdict can be multiplied by considering ways of creating added structural complexity other than by just adding negation. For example, the intuited felicity of (23a) and (23b) indicates that (19) can also carry the implicatures $\neg$ Basic(24a) and $\neg$ Basic(24b). In this case, since (24a) and (24b) are themselves compatible with one another, it would also be possible to deny both implicatures at the same time.

- (23) a. No, the detective also concluded that Jed did not steal the jewels unless they were hidden in the books!
b. No, the detective also concluded that either Jed stole all of the jewels or the jewels are kept in a different apartment!
- (24) a. The detective concluded that Jed did not steal the jewels unless they were hidden in the books.
b. The detective concluded that either Jed stole all of the jewels or the jewels are kept in a different apartment.

³The example in Fox and Katzir (2011) that our example (19) adapts, given in (i), involves association of focus with *only*. Fox and Katzir note that in the context of the first sentence in (i), the second sentence can implicate $\neg$ Basic(ii), despite the fact that (ii) is more complex, crucially due to the presence of negation.

- (i) Detective B concluded that the robbers [stole the books but not the jewelry]_F.
Detective A only concluded that the robbers [stole the books]_F.
(ii) Detective A concluded that the robbers stole the books but not the jewelry.

Fox and Katzir use this observation to argue that more complex alternatives can be excluded by *only* if they arise from substitutions of material that is *explicitly provided in the context*, here given by the first sentence of (i) (see also Section 3.3 below). However, while their implicit claim is that such contextual support is *necessary* to enable the exclusion of a more complex alternative, they do not actually consider controls where the relevant sort of contextual support is missing. In fact, aligned with our intuitions about (19), observations in Hirsch (2024) suggest that such support is not needed in the case at hand (cf. also Hirsch and Schwarz 2023): even without being preceded by the first sentence in (i), the second sentence could well be taken to imply that Detective A did *not* conclude that the robbers did not steal the jewelry.

What is more, it seems that (19) may well carry further implicatures, based on alternatives that remove or add negation from the embedded clauses in (24), which is again made possible by the fact that *consider* removes symmetry. But the intuited availability of $\neg$ Basic(24a) and $\neg$ Basic(24b) themselves suffices to support point we wish to make here. Since both (24a) and (24b) are transparently more complex than (19), and since it seems evident that this holds for any sentences with the same basic meanings, these implicatures are again incorrectly prevented by CF.

At this point, it is important to acknowledge that there is friction between the observations that we have presented here as challenges to CF and closely related observations reported in previous literature. It seems that the strengthening observed for sentences with embedding under a symmetry removing predicate sometimes *is* constrained in a way that would CF lead us to expect. Fox and Spector (2018: 15) discuss the example in (25), a case of embedding under the modal necessity operator *require*, which removes symmetry in the same sense as *conclude*.

(25) You're required to buy some of the furniture.

For (25), Fox and Spector describe a pattern that is different from the one we saw for (19). Contrasting with the reply to (19) in (22b), the reply to (25) in (26) seems infelicitous. This suggests that (25) cannot be understood as carrying the implicature $\neg$ Basic(27), that we are not required to not buy all of the furniture, or, equivalently, that we are allowed to buy all of it. Given the structural complexity of the potential alternative in (27), this is as expected under CF.

(26) #No, we are not allowed to buy all of it!

(27) You're required to not buy all of the furniture.

How is one to reconcile Fox and Spector's observation about (25) with our observation about (19)? Recall that the judgments we reported for (19) were about its use in (21), where (19) is placed in a context intended to ensure that the potential alternatives of interest are contextually relevant. For a proper comparison, we should therefore do the same for Fox and Spector's example (25). Consider, then, the exchange in (28), where (25) serves as a response to an utterance that plausibly establishes the relevance of the alternative in (27).

(28) A: Did you listen to the instructions? Tell me exactly what we have to do.

B: We're required to buy some of the furniture.

We should now judge the felicity of the denial in (26) as a continuation of (28), perhaps again as uttered by a third person. We submit that the added context in (28) indeed renders (26) more natural. Note, moreover, that the felicitous denial of (19) in (22b) features the additive particle *also*. Additive particles are often obligatory when they are permitted (e.g. Bade 2016). Given this, we might expect that the acceptability of a denial of $\neg$ Basic(25) can be improved by adding *also*, as in (29), and this seems to be so.

(29) No, we are also not allowed to buy all of it!

We find the relevant sort of implicature even more clearly available for the variant of Fox and Spector's example (25) given in (30). Confirming the presence of the implicature $\neg$ Basic(33), the denial in (32) seems quite unexceptional as a conceivable response to the use of (30) in the exchange in (31). In this case, the reason for why the truth or falsity of the implicature matters seems

particularly easy to accommodate, and this might be what renders the denial of the implicature particularly natural.

- (30) We have to press some of the buttons.
- (31) A: Did you listen to the instructions? Tell me exactly how to diffuse the bomb.
B: We have to press some of the buttons.
- (32) No, we also must not press all of them!
- (33) We have to not press all of the buttons.

Given these observations, we suggest that Fox and Spector's observation that the denial of (25) in (26) can be judged infelicitous should not be interpreted as evidence for CF. We propose that this observation should instead be chalked up to the condition that the members of Excl be contextually relevant.

To further strengthen the empirical basis for our evaluation of CF, let's look at one additional example. Sentence (34) again features a necessity modal, here the auxiliary *must*, whose modal base is restricted by an initial purpose clause. In parallel to what we reported above for the case with symmetry removing *consider* in (19), both (35a) and (35b) are felicitous responses, indicating that (34) can be understood as implying either $\neg$ Basic(36a) or $\neg$ Basic(36b). As in the case of (19), these implicatures can both be drawn at the same time without contradiction, since *must*, in virtue of expressing necessity, again removes symmetry.

- (34) To get a cookie, she must eat some of the plums.
- (35) a. No, she must eat all of them!
b. No, she also must not eat all of them!
- (36) a. She must eat all of the plums.
b. She must not eat all of the plums.

We moreover note that, still paralleling the case of (19), the implicature $\neg$ Basic(36b) is available despite the fact that the alternative in (36b), like presumably any sentence with the same basic meaning, is more complex than (34) itself, so that CF prevents this meaning from being a member of Excl. And again, we can multiply felicitous denials that challenge CF by considering ways of adding complexity other than just adding negation. Each of the responses in (37) is unquestionably felicitous, indicating that (34) can implicate the negation of the basic meaning of each of the alternatives in (38), despite the fact that each of them is more complex than (34) itself. In fact, it seems natural to interpret (34) as conveying that eating some of the plums is not just a necessary condition for getting a cookie, as entailed by Basic(34), but also a *sufficient* condition. This suggests that (34) implicates the negation of *any* conceivable clausal argument of *must*, regardless of its linguistic form.

- (37) a. No, she also must not sing *Hallelujah* more than once!
b. No, she also must play the piano for at least one hour!
c. No, she also must either eat an orange or take a walk!
- (38) a. She must not sing *Hallelujah* more than once.
b. She must play the piano for at least one hour.

- c. She must either eat an orange or take a walk.

It is notable that the relevant intuitions about (34) seem crisp even in the absence of the sort of context provided by A's utterance in the discourses in (21), (28), and (31). We interpret this as showing that the initial purposes clause in (34) can serve to establish the relevance of the participating alternatives, much as A's utterance is intended to in (21), (28), and (31).

In sum, the observations presented above challenge Katzir and Fox's proposal to solve the symmetry problem through a *general* constraint that bars certain propositions from Excl in virtue of the necessary structural complexity of their linguistic encoding. In basic cases, CF has the intended effect, by preventing one member of a pair of symmetric alternatives from contributing its meaning to Excl, correctly preventing an implicature that excludes that member. However, the complexity filter being a general constraint, it predicts observable restrictions on the membership of Excl even in situations where the presence of an operator removes the symmetry that would otherwise obtain. We have argued that this prediction is incorrect, and we in fact saw reasons to doubt that Excl is constrained in any way by how its members can be linguistically encoded.

Before further evaluating CF, we note that the data introduced here as a challenge to CF pose no threat to the blocking constraint BC. To see this, it is useful to first review that for (16), BC prevents the unavailable strengthening in virtue of the specific way that (16) relates to (17a,b), viz. the symmetry of (17a,b) relative to (16). It is in virtue of this symmetry that the strengthened meaning that exhaustification of (16) outputs relative to the exclusion set $\{\text{Basic}(17b)\}$, $\text{Basic}(16) \wedge \neg \text{Basic}(17b)$, is equivalent to $\text{Basic}(17a)$. In turn, with (17a) being no more complex than (16), it is in virtue of this equivalence that BC prevents (16) from having the strengthened meaning $\text{Basic}(16) \wedge \neg \text{Basic}(17b)$.

Now, crucially, due to the embedding under symmetry-breaking *conclude*, no parallel relation holds between the sentences in (19) and (20a,b). Exhaustification of (19) relative to the exclusion set $\{(20b)\}$ yields $\text{Basic}(19) \wedge \neg \text{Basic}(20b)$, the proposition that the detective concluded that Jed stole some of the books, but did not conclude that Jed did not steal all of the books. This proposition is *not* equal to $\text{Basic}(20a)$, the proposition that the detective concluded that Jed stole all of the books. So at least (20a) is *not* a candidate for blocking exhaustification of (19) relative to $\{(20b)\}$. Moreover, while $\text{Basic}(19) \wedge \neg \text{Basic}(20b)$ is the basic meaning of, say, sentence (39), this sentence, like arguably any other sentences with the same basic meaning, is more complex than (19). If so, then the strengthened meaning $\text{Basic}(19) \wedge \neg \text{Basic}(20b)$ is not blocked, hence is permitted by BC.

- (39) The detective concluded that Jed stole some of the books, but
the detective did not conclude that Jed did not steal all of the books.

This establishes that BC correctly allows for (19) to convey the implicature $\neg \text{Basic}(20b)$, that the detective did not conclude that Jed did not steal all of the books. In much the same way, it can be demonstrated that BC likewise permits any of the other specific implicatures for (19) diagnosed above, $\neg \text{Basic}(20a)$, $\neg \text{Basic}(24a)$, and $\neg \text{Basic}(24b)$. Each of these implicatures is permitted on the grounds of yielding a strengthened meaning that without strengthening would require such a complex linguistic encoding that it evades blocking under BC. In fact, for the same reason, BC correctly allows for (19) to have a strengthened meaning that entails the conjunction of all of these implicatures. Even more broadly, for any proposition p, BC allows for (19) to implicate the

statement that the detective did not conclude p , as well as the conjunction of any number of such statements. For all we have seen, this is as it should be, keeping in mind that which implicatures are actually observed in a given utterance context is assumed to depend on relevance.

To recap again, we saw in this section that CF cannot be a general constraint on the alternatives in Excl, since this would preclude certain implicatures that are attested in cases in which an embedding environment removes symmetry between alternatives of different complexity. We also saw that the relevant implicatures are correctly permitted under the blocking constraint BC. Under BC, in contrast to CF, the addition of a symmetry-breaking operator in pair of alternatives that would otherwise be symmetric is expected to allow for both alternatives to yield an implicature. In other words, for any given pair of alternatives, it is only possible for one alternative to block exhaustification via exclusion the other if the two alternatives are symmetric.

To be sure, instead of viewing CF as a general constraint on alternatives, one could stipulate that CF only applies in situations where alternative are symmetric to each other, as a form of tie breaker (see Haslinger and Schmitt 2024 for a recent response long these lines). But why would CF apply in those cases, and not more generally? It seems to us that much of the attractiveness of CF as a solution to the symmetry problem derives from its predictive power outside the cases of symmetry.

In the next section, we argue that apart from being overly restrictive, CF also does not actually solve the symmetry problem in full. We will see evidence for cases of symmetry in which CF cannot explain why certain implicatures are available and others unavailable, since the problematic formal alternatives are no more complex than what is asserted. As we will see, it correctly predicts the absence of certain implicatures even when the expressions encoding the excluded alternative meaning are not complex.

3 The case for blocking

CF prevents the unwanted strengthening in (8) for (1) by exploiting a limitation of English relative to the structure of (1). In contrast to ALL, $\neg$ ALL is not lexicalized. Expressing $\neg$ ALL therefore requires a syntax that transparently encodes negation, but adding negation to (1) would increase complexity. It is in virtue of this increase that CF here has the intended effect. Notably, then, if the meaning $\neg$ ALL were lexicalized, then CF would do nothing to resolve symmetry in this case.

Exploring this line of thought, we will now present a challenge to CF as a general solution to the symmetry problem. We will discuss cases where unwanted meanings are carried by acceptable sentences which lack the complexity that would be needed for CF to screen them out. We refer to the such sentences as *simplex threats*.

Simplex threats suggest that CF is not sufficient as a solution to symmetry resolution. By the same token, they furnish potential arguments for the presence of BC as a constraint on exhaustification. In each case to be discussed, the unwanted meaning that the simplex threat supports, while potentially output by Exh, is correctly blocked by BC. Of course, it could be that a pragmatic constraint other than BC is at play. We already saw that quantity implicatures appear to be constrained by what counts as a plausible level of precision for an answer in a particular context. As we will see, though, similar explanations for apparent cases of blocking are at least not easily conceivable.

Our argument for a blocking condition is based on three types simplex threats, each of the three case studies crucially builds on previous work. The simplex threat based on expressed negation (Section 3.1) is a variant of a case discussed in Romoli (2013). Like we do for our case, Romoli interpreted his example as a challenge for CF. The simplex threats with antonyms (Section 3.2) are related to data in Breheny et al. (2018), and directly inspired by cases in Buccola et al. (2022). Again, like we do for our variants, Breheny et al. and Buccola et al. used their data to challenge CF. Finally, the case of contextually supported simplex threats (Section 3.3) is centrally discussed in Katzir (2007) and Fox and Katzir (2011). We will discuss their additional assumptions aimed at diffusing this simplex threat (and potentially the others as well) in Section 4.

3.1 Simplex threat 1: enabled by negation

The first type of simplex threat we present arises in certain data where an overt negation interacts scopally with another operator (cf. Romoli 2013). Here we focus on a case where negation is interpreted in the scope of an existential indefinite, as in (40).⁴

(40) Some players have not responded.

The basic meaning of this sentence can be given in (41) (where D is the domain of individuals). As stated in (42), we observe an asymmetry in the intuitively available strengthenings much like the one that we described for (1). Sentence (40) can be understood as conveying that some but not all players responded, see (42a), but not as conveying that none of them did, see (42b).

(41) Basic meaning of (40):
 $SOME(P)(D-R)$

(42) a. Available strengthened meaning of (40):
 $SOME(P)(D-R) \wedge \neg ALL(P)(D-R) = \neg ALL(P)(R) \wedge SOME(P)(R)$
 b. Unavailable strengthened meaning of (40):
 $SOME(P)(D-R) \wedge \neg SOME(P)(R) = NO(P)(R)$

CF does not derive this asymmetry. Sentence (43a) below, which permits a surface scope reading with the basic meaning $NO(P)(R)$, is no more complex than (40). Therefore, since (42a) is the strengthened meaning of (40) that can result from, say, the exclusion set $\{NO(P)(R)\}$, CF correctly permits this strengthening. However, sentence (43b), with the basic meaning $SOME(P)(R)$, is not more complex than (40), either. Therefore, since (42b) is the strengthening for (40) that would result from, say, the exclusion set $\{SOME(P)(R)\}$, CF permits this unwanted strengthening as well. The sentence in (43b), then, is a simplex threat.

(43) a. All players have not responded.
 b. Some players have responded.

If CF does not capture the asymmetry in (42), then what does? We propose that the blocking condition BC has the intended effect. BC correctly prevents (42b) as a strengthened meaning for

⁴Sentence (40) is variant of an example discussed in Romoli (2013), which centrally features a negated universal, as in *Not all players have responded*. Romoli's case could be captured by supplementing CF with an additional constraint, the *Atomicity Constraint* proposed in Trinh and Haida (2015). We do not have space to discuss this constraint here, but we note that its intended effect in any case does not extend to our example in (40).

(40). It is blocked due to the existence of a sentence that is no more complex than (40) and that has (42b) as its basic meaning. For example, (43a) is such a sentence. We moreover argue that, in contrast, BC permits the strengthened meaning in (42a). To be sure, there are sentences that have (42a) as their basic meaning. Example (44) is a case in point. However, (44) is more complex than (40). We submit that this is in fact true for all sentences with the basic meaning (42a). If so, then BC correctly permits (42a) as a strengthened meaning for (40).

(44) Some but not all players have responded.

In sum, the pattern of strengthening with sentence (40) cannot be captured by CF without further assumptions, while BC makes the correct prediction. The crucial feature of example (40) that distinguishes it from our original case in (1) is that it already contains negation. It is on those grounds that a symmetric pair of linguistic alternatives can be derived by removing rather than adding complexity. We now turn to a case in which simplex threats arise by simple substitutions, based on antonymic content vocabulary.

3.2 Simplex threat 2: enabled by antonyms

Another type of simplex threat can arise in cases where a predicate that appears in the scope of another operator has a lexical antonym (cf. Breheny et al. 2018, Buccola et al. 2022). Here we consider the case where a predicate of this sort appears in the scope of an existential indefinite.

(45) Some players were inside.

The basic meaning of (45) is given in (46). As stated in (47), we once again observe an asymmetry in the intuitively available strengthenings. (45) can be understood as conveying that some but not all players were inside, see (47a), but not as conveying that all were inside, see (47b).

(46) Basic meaning of (45):
 $SOME(P)(In)$

(47) a. Available strengthened meaning of (45):
 $SOME(P)(In) \wedge \neg ALL(P)(In)$
 b. Unavailable strengthened meaning of (45):
 $SOME(P)(In) \wedge \neg \neg ALL(P)(In) = ALL(P)(In)$

It may initially appear as though CF captured this asymmetry, as CF may seem to apply to (45) in the same way it applies to the basic example (1). To begin, since sentence (48a), with basic meaning $ALL(P)(In)$, is no more complex than (45), CF permits the exclusion set to be, for example, $\{ALL(P)(In)\}$. Hence it permits exhaustification that yields the available strengthened meaning in (47a). At first sight, CF seems to also correctly prevent the undesirable inference. After all, while sentence (48b) transparently has $\neg ALL(P)(In)$ as its basic meaning, the presence of syntactic negation renders (48b) more complex than (45).

(48) a. All players were inside.
 b. Not all players were inside.

However, closer inspection reveals CF to be insufficient to prevent the exclusion set from containing $\neg ALL(P)(In)$. The reason is that this same proposition is also expressed by a different sentence,

one that crucially is no more complex than (45), viz. (49), where *inside* is replaced by its antonym *outside*:

(49) Some players were outside.

Sentence (49) transparently expresses the proposition $\text{SOME}(P)(\text{Out})$. Since the complementarity of the antonyms *inside* and *outside* guarantees the equality of the two sets Out and D–In, this basic meaning is equivalent to $\text{SOME}(P)(\text{D–In})$ and hence to $\neg\text{ALL}(P)(\text{In})$. Therefore, CF does not prevent an exclusion set containing this proposition after all. Allowing for, say, $\{\neg\text{ALL}(P)(\text{In})\}$ as the exclusion set, CF after all fails to prevent the unavailable strengthened meaning in (47b). In other words, sentence (49) is a simplex threat.

The blocking condition BC captures the asymmetry in (47). BC correctly blocks (47b) as a strengthened meaning for (45), due to the existence of a sentence like (48a), which is no more complex than (45) and has (47b) as its basic meaning. While there are sentences with the basic meaning (47a), we submit that they all are more complex than (45). For example, both of the sentences in (50) have (47a) as their basic meaning. But given their complexity relative to (45), their existence does not stand in the way of (45) having this same meaning as a result of strengthening. We submit that this extends to all sentences with the basic meaning (47a). If so, then BC correctly allows for (45) to acquire this meaning from strengthening.

- (50) a. Some but not all players were inside.
b. Some players were inside and some were outside.

The example in (45) illustrates that symmetric linguistic alternatives can evade the grasp of CF in virtue of featuring predicates that have antonyms. By introducing semantic negation without adding syntactic complexity, replacing such a predicate with its antonym can give rise to a simplex threat. We will next attend to yet another type of simplex threat, cases where a problematic alternative should be permitted in virtue of features of the linguistic context.

3.3 Simplex threat 3: enabled by context

Under a construal of the notion of syntactic complexity that seems initially plausible and that we have tacitly assumed so far, a structure β counts as no more complex than a structure α only if β can be obtained from α by merely deleting constituents of α or replacing terminal nodes. However, based on an observation in Matsumoto (1995), Katzir (2007) suggests that for the purposes of the theory of exhaustification, this condition is too demanding. Consider the conjunctive sentence given by the concatenation of the clause in (51a) with (51b).

- (51) a. It was warm yesterday ...
b. ... and it is a little bit more than warm today.

While the basic meaning of (51a) merely implies that it was warm yesterday, Matsumoto observed that as part of the conjunctive sentence, (51a) intuitively permits a strengthened meaning that implies in addition that yesterday it was *not* a little bit more than warm. Hence strengthening can exclude the proposition expressed by the linguistic alternative to (51a) stated in (52).

- (52) It was a little bit more than warm yesterday ...

Under CF, this is surprising since (52) appears to have more structure than (51a). In fact any sentence with the same basic meaning as (52) seems to have more structure than (51a). This leads Matsumoto to conclude that a complexity constraint on the exclusion set is not viable. However, Katzir suggests that Matsumoto's observation can be reconciled with CF under a suitable construal of the notion of complexity that CF is taken to reference. Katzir proposes that for the purposes of CF, a constituent that appears in the linguistic context of an utterance is treated like a lexical item, in the sense that CF applies *as though* this constituent lacked internal structure. Hence in the case at hand, given that the constituent *a little bit more than warm* appears in the linguistic context of (51a), substituting it for *warm* will not actually be taken to incur an increase of complexity for the purposes of CF. If so, then (52) counts as no more complex than (51a), reconciling CF with the observed strengthening.

Assuming the amended understanding of complexity motivated by Matsumoto's example, consider now the sentence given by the concatenation of (53a) with (53b), which adapts similar examples discussed in Katzir (2007) and Fox and Katzir (2011).

- (53) a. Last week, some of them responded, ...
 b. ... and this week (too), not all of them responded.

We take it that, with *them* referring to the players, (53) can give rise to the very same strengthening asymmetry that we initially detected for sentence (1). Even in the context of the continuation (53b), (53a) can be understood as conveying that some but not all players responded last week, as in (54a). This meaning for (53a) can be brought out clearly by including the additive particle *too* in (53b), thereby forcing the strengthened interpretation for (53a) in (54a). In contrast, as indicated in (54b), with or without the additive particle, (53a) certainly cannot be understood as conveying that last week all of the players responded.

- (54) a. Available strengthened meaning of (53a):
 $\text{SOME}(\text{P})(\text{R}) \wedge \neg \text{ALL}(\text{P})(\text{R})$
 b. Unavailable strengthened meaning of (53a):
 $\text{SOME}(\text{P})(\text{R}) \wedge \neg \neg \text{ALL}(\text{P})(\text{R}) = \text{ALL}(\text{P})(\text{R})$

Under the amended construal of complexity, *both* of the linguistic alternatives to (53a) in (55) qualify as no more complex than (53a). In particular, this is true for (55b), on the grounds that the constituent *not all of them responded* appears in the linguistic context (53b). CF therefore permits the unwanted strengthening in (54b) alongside the attested one in (54a). Hence the alternative (55b) constitutes yet another case of a simplex threat.

- (55) a. Last week, all of them responded ...
 b. Last week, not all of them responded ...

On the other hand, the asymmetry in (54) is once again captured by BC. BC applies to this case in the very same way as it does to our initial example in (1). The unavailable strengthening in (54b) is blocked by sentence (55a), for example, given that (55a) has (54b) as its basic meaning. The available strengthening in (54a) correctly evades blocking. While sentence (56) has (54a) as its basic meaning, it is more complex than (53a). Assuming that this is true for all sentences with this basic meaning, BC correctly permits the strengthening in (54a).

(56) Last week, some but not all of them responded . . .

To recap, we have now seen three types of simplex threats, cases where CF fails to exclude alternatives that would yield unavailable strengthenings. We also saw that in each case, the unavailable strengthening is correctly blocked under BC. In the absence of an alternative explanation for why the potential strengthenings in question are unavailable, this furnishes an argument that BC is needed as a constraint on the output of exhaustification.

How conclusive is this argument? We will next scrutinize a possible alternative approach to the simplex threats that we presented, emerging from proposals in Katzir (2007) and Fox and Katzir (2011). We will argue that the alternative account does not in fact successfully remove the challenge posed by simplex threats. Moreover, we will argue that a crucial assumption that the alternative is based on, which we will call the *stalemate assumption*, cannot be maintained.

4 Symmetric stalemate to the rescue?

Our argument for the blocking condition BC based on simplex threats could in principle be called into question by motivating further constraints on the exclusion set. It is conceivable that such further constraints could pave the way to an alternate account of our simplex threat data. We will now discuss one attempt to supplement CF in this way, viz. a cluster of assumptions motivated in Katzir (2007) and Fox and Katzir (2011).

Katzir and Fox propose that in the grammar of exhaustification, CF operates alongside another central principle that regulates the membership of Excl. The principle holds that in cases where grammatical constraints on linguistic alternatives permit two alternatives that are symmetric, so that excluding both would yield a contradiction, it is not possible, regardless of context, to simply selectively omit one of the two alternatives and exclude just the other. In other words, the claim is that symmetry permitted by CF leads to a stalemate between mutually contradictory inferences, a stalemate that context is assumed to be unable to break.⁵ For reference, we state the stalemate assumption as in (57).

(57) Stalemate assumption:
Symmetric alternatives permitted by CF cannot be selectively omitted from Excl

Naturally, when CF is supplemented with the stalemate assumption, the predictions for our simplex threat data change. The central feature of those data is precisely that there are at least two symmetric alternatives which satisfy CF. The prediction is now that in the presence of symmetric alternatives, *none* of the the potential inferences will be available. Concretely, as matters stand, in each of (42), (47), and (54), the stalemate assumption prevents the unavailable strengthening in (b). Of course, it also rules out the available implicature in (a) in each case, but Fox and Katzir (2011) illustrate how it can still help as part of a strategy to address such simplex threats.

⁵If one assumes with Fox (2007) that exclusives in fact ignore alternatives that are not ‘innocently excludable,’ then symmetric alternatives will simply be ignored at the point of exhaustification. In our terms, they are never part of Excl.

4.1 The stalemate assumption falls short

Let us look at our third simplex threat case, the case invoking the linguistic context, for which Fox and Katzir (2011) expressly appeal to the stalemate assumption to avoid the undesirable implicature. To see how this helps, consider again the coordinated sentence given by (58a) and (58b), repeated from (53), as well as the two possible alternatives to (58a) in (59a) and (59b), repeated from (55).

- (58) a. Last week, some of them responded, ...
b. ... and this week (too), not all of them responded.
- (59) a. Last week, all of them responded ...
b. Last week, not all of them responded ...

Recall that, given the baseline data discussed in Matsumoto (1995), the contextual support from (58b) should allow for (59b) to serve as an alternative to (58a), deriving an unattested strengthening. The stalemate assumption, however, ensures that if both alternatives are available, *neither* inference can be derived. Refining the assumptions about the effect of context, Fox and Katzir (2011: fn. 16) suggest that including the meanings of contextually provided complex alternatives in the exclusion set is merely optional. This refinement has the intended effect for (58a). If we do not include (59b) in Excl, exhaustification can output the strengthened meaning in (54a); if we include both, neither implicature is derived.

But can this strategy be extended to the other two types of simplex threats that we have seen? Once we assume the stalemate assumption, the initial prediction for our other simplex threats is again that we should see no strengthening. While the problematic strengthening based on the simplex threat is now correctly ruled out, how can we permit the strengthening that *is* available?

A successful strategy would again have to be to somehow eliminate the undesirable alternative. A further assumption advanced in Fox and Katzir (2011) opens up one possible route. Fox and Katzir propose that in addition to being constrained by CF, the form of linguistic alternatives is shaped by F(ocus)-marking. The possible alternatives to a given sentence are assumed to differ in this sentence only with regard to F-marked constituents. Put differently, modulo F-marked constituents, an alternative to a given sentence is required to be an exact copy of that sentence. Certain alternatives can then be avoided by assuming a particular F-placement. To show this, we return to the relevant examples (40) and (45), repeated in (60a) and (60b), respectively.

- (60) a. Some players have not responded.
b. Some players were inside.

For sentence (60a), the problematic pair of symmetric alternatives is stated again in (61a), repeated from (43). Likewise, the relevant symmetric alternatives to (60b) are stated in (61b), repeated from (48a) and (49).

- (61) a. (i) All players have not responded.
 (ii) Some players have responded.
b. (i) All players were inside.
 (ii) Some players were outside.

Suppose now that the sentences in (60) carry F-marking that includes *some* in an F-marked constituent, but not the negation *not* or the antonymic predicate *inside*. For example, suppose for concreteness that both structures only feature narrow F-marking on *some*, as in (62).

- (62) a. [Some]_F players have not responded.
 b. [Some]_F players were inside.

Under Fox and Katzir's proposal, such F-marking ensures that that alternatives can differ from those sentences in the position of *some* but nowhere else. Therefore, while the sentences in (61a-i) and (61b-i) qualify as alternatives to (62a) and (62b), respectively, the simplex threats in (61a-ii) and (61b-ii) do not. With the simplex threats so eliminated by grammar, the strengthenings in (42a) and (47a) are correctly predicted to be available. That is, it is correctly predicted that (60a) can be understood as excluding (61a-i), to convey that some but not all players have responded, and (60b) as excluding (61b-i), to convey that some but not all players were inside.

However, Fox and Katzir's proposal also makes predictions about the sentences in (60) that are not in fact consistent with intuitions. The account is too permissive in some ways, and too restrictive in others.

On the one hand, the proposal allows for the potential strengthenings in (42b) and (47b) to arise under conditions that support structures with appropriate F-marking, viz. structures where *not* and *inside* are included in F-marked constituents, while *some* is not. For example, (42b) and (47b) would be expected to arise from the structures in (63), with narrow focus on *not* and *inside*. Since there is in fact no reason to expect that such F-marking is categorically excluded, it is predicted that the strengthenings in (42b) and (47b) can be observed under certain conditions. However, as implied by our initial characterization of the data, intuitions are clear that these strengthenings are not in fact available under any conditions. In fact, even with a prosody favoring the F-marking in (62), the possibility of the implicature based on substituting *some* persists. It seems that the assumption that scalar implicatures are constrained by limiting substitutions to F-marked content is not in fact supported by the evidence.

- (63) a. Some players have [not]_F responded.
 b. Some players were [inside]_F.

On the other hand, Fox and Katzir's proposal also leads one to expect that the availability of the strengthenings in (42a) and (47a) requires a particular F-marking. Specifically, strengthening should be obviated by F-marking that leads to a symmetric stalemate. Under conditions that force both *some* and *not* to be included in an F-marked constituent in (60a), both (61a-i) and (61a-ii) would qualify as alternatives to (60a). Likewise, both (61b-i) and (61b-ii) would qualify as alternatives to (60b) under conditions that force both *some* and *inside* to be included in an F-marked constituent in (60b). Under such conditions, a stalemate should ensue, and hence we should not see either of the strengthenings in (42) and (47), respectively. It is predicted, for example, that neither strengthening is observed under conditions that force narrow focus on *some* and *not* or *inside*, as in (64), or that force broad focus on the entire sentence, as in (65).

- (64) a. [Some]_F players have [not]_F responded.
 b. [Some]_F players were [inside]_F.
 (65) a. [Some players have not responded]_F.

- b. [Some players were inside]_F.

This prediction too seems incompatible with intuitions. As hard as one may try to force the sort of F-marking shown in (64) and (65), it seems that the strengthening asymmetries described in (42) and (47) persist, with (42a) and (47a) continuing to be available. In particular, these asymmetries are still intuited when the sentences in (60) are produced with prosodic focus on *some* and *not* or *inside*, in an attempt to force the double narrow F-markings in (64). And, paralleling an observation in Romoli (2013), they also persist in responses to a wh-question like *What happened?*, which should favor the broad F-markings in (65).

In sum, the strategy of appealing to the stalemate assumption and F-marking to avoid the simplex threat fails on two counts: First, we showed direct counter-evidence to the proposal that the exclusion set is constrained by F: focus placement that should make undesirable alternatives available and desirable ones unavailable do not change which implicature is available in cases of symmetry. The computation of implicatures is not in fact constrained to substitutions motivated by focus and F-marking. An alternative explanation is needed, and we propose that the blocking condition BC can fill this gap.

The second reason that the strategy fails is that focus marking that should make *both* alternatives available fails to lead to a stalemate. These examples cast serious doubt on the validity of the stalemate assumption itself.

Much of the recent literature on the symmetry problem assumes the stalemate assumption as given. Under this assumption, a solution to the symmetry problem amounts to a way of preempting a stalemate between symmetric alternatives that would otherwise be expected, breaking the tie between two individually conceivable but mutually incompatible implicatures. And yet, aside from its role in fending off apparent counterexamples to CF, we have not seen any reason yet for why the stalemate assumption is actually needed, and we have now encountered apparent countevidence.

If the formal stalemate assumption is incorrect, this would change the perspective on the nature of symmetry problem. The task would no longer be to explain how, in a case of symmetry, symmetry between alternative can be ‘broken’ in order make certain implicatures available that should be impossible due to a stalemate. Instead, the task is now to explain why any of the potential mutually incompatible implicatures should ever be unavailable. We have seen arguments that some implicatures are unavailable due to blocking, and others because they would not contribute relevant information. Neither type of case required appeal to the stalemate assumption.

There is another set of cases though, where the stalemate assumptions seems crucial. The stalemate assumption could be, and has been, invoked to explain certain cases of symmetry where no strengthening in any direction is observed. It seems that for such cases, an explanation based on some sort of stalemate between symmetric alternatives is inescapable. In fact, we suspect that such cases may have served as the original motivation for the stalemate assumption. We will discuss such cases next, concluding that they indeed motivate a notion of stalemate between symmetric alternatives—but that the stalemates in question can be understood in pragmatic terms alone.

4.2 Pragmatic stalemates

We characterized the symmetry problem as involving cases in which only a subset of several conceivable but mutually exclusive implicatures are intuitively available. However, there are also cases of symmetry in which *none* of several conceivable but contradictory implicatures appear to be available. By giving up the stalemate assumption, we seem to have created a new problem: How can we explain such empirically attested stalemates? Let's look one such case of a 'de facto' stalemate:

- (66) a. A: I have a yellow bicycle.
b. B: # That's not true! You have a cheap bicycle.

Here B's objection seems contrived, but why? The sentence should have the following two alternatives, and—unless we make the stalemate assumption—shouldn't we now expect that an implicature based on either of them is viable?

- (67) a. Alternative 1: I have an expensive bicycle
b. Alternative 2: I have a cheap bicycle

And yet appealing to a formal stalemate condition to explain (66) seems unnecessary. There is a plausible pragmatic reason for why A cannot be taken to court by B for having implied that their bicycle is not cheap. If A arbitrarily adds the alternative *expensive* to the exclusion set, this is not reliably recoverable by B, and hence B is not entitled to object to the implications of this choice on the grounds of what A asserted. Of course A would not even try this, knowing that their message would not be recoverable, and resort to a different way to convey this meaning instead. In other words, such implicatures are intuitively non-recoverable from the utterance.⁶

The pragmatic account for empirically attested stalemates sketched builds on the insights from the formal stalemate condition: Fox and Katzir (2011) also aim to rule out arbitrary enrichments as the ones discussed here, but they attribute this to the stalemate condition on formal alternatives, and propose to restrict Exh to only apply to 'innocently excludable' alternatives. Katzir (2013) applies this strategy to rule out certain alternatives to modifying adjectives in cases like (66). We submit that the impossibility of arbitrary and hence non-recoverable inferences is instead a pragmatic effect that can be explained without appeal to these assumptions.

This pragmatic account of stalemates predicts that context should in principle be able to make an implicature recoverable, despite of the availability of symmetric alternatives, a state of affairs which Fox and Katzir (2011) argue is impossible. We present the following pair of examples as evidence to the contrary. Consider the dialogues in (68) and (69):

- (68) What do you do to protect your teeth?
Kid 1: I brush my teeth. (And) I don't eat candy.
Kid 2: I brush my teeth (too).
(69) What do you do to protect your teeth?
Kid 1: I brush my teeth. (But) I don't floss.

⁶We note that B's response in (66) would be fine in a context in which (for some contrived reason) everyone knows that for a bike, being yellow implies being expensive. At first blush this seems to contradict accounts like the stalemate assumption that formally prohibit this implicature. But in a context in which this inference holds, one arguably does not even need an implicature to license the required inference that makes the dialogue in (66) coherent.

Kid 2: I brush my teeth (too).

The dialogues are fully parallel in linguistic form, but nevertheless differ with regard to the direction of symmetry resolution. What we are interested in are the inferences that can be drawn from Kid 2's assertion in each case.

Let's start with (68), which is based on a similar example in Breheny et al. (2018). Kid 2 can be taken in this dialogue to imply that they *eat candy*, but not that they *don't eat candy*. Breheny et al. (2018) point to cases like (68) as a challenge to CF. In our terms, the problem is the presence of a simplex threat. The threat is from the positive alternative *I eat candy*, which is symmetric to the alternative actually excluded in (68), *I don't eat candy*.

Whatever explains this example, it seems to us that it will not be a more refined formal filter on alternatives. To see why, we need to compare examples like (68) with examples like (69). Here, we intuit that Kid 2 in (69) can be understood as implying that they *do not floss* either—apparently excluding the *positive* alternative *I floss*, and not the negated alternative *I don't floss*. Dialogues (68) and (69) contrast not because how the alternatives are, or can be, linguistically encoded, but because of what we know about how flossing and eating candy affect our teeth. There appear to be two symmetric alternatives that are in principle available, but symmetry is resolved in opposite directions depending on world knowledge.

Fortunately, a pragmatic explanation of the difference is straightforward: The questions in (68) and (69) makes reporting *not eating candy* and *flossing* directly relevant but the same is not true of their negations. The assumptions that Excl contains only relevant propositions then suffices to capture the observed pattern of exhaustification.

Taken together, then, (68) and (69) seem to show that both negated and unnegated alternative are in principle available, and that context *can* break the stalemate between these symmetric alternatives, providing further support for the pragmatic account of stalemates.⁷

5 Apparent counterexamples

We have seen several arguments that speak against the account in terms of CF, and in favor of an account in terms of BC. We now turn examples that seem incompatible not just with CF (and the

⁷We note that example (69) is parallel in structure to examples discussed in Section 3.3, such as (58). More complex alternative like the negated alternative *I don't floss* are actually assumed to be allowed by CF if they are contextually mentioned in the context. So the prediction of the CF-account is that no implicature should be viable here due to the stalemate assumption. As noted above, Fox and Katzir (2011) propose to avert this problem by assuming that the contextually mentioned negative alternative can optionally be omitted from Excl. So example (69) is compatible with CF if we assume that the speaker chose the option to ignore the contextually provided negative alternative. What about (68)? What one would need to say here is a refined formal filter that keeps the more complex negated alternative and discards the simpler, unnegated alternative. For proposals along those lines see Trinh (2018) and Haslinger and Schmitt (2024). The reason both symmetric implicatures are in principle possible under this account is that taking the contextually provided alternatives into account will force the implicature in (68) (because the more complex negated alternative is ruled out by CF), while not taking it into account will force the implicature in (69) (because the additional assumptions make the negated alternative the preferable one then). Note that relevance plays no role in this potential formal account. While technically compatible with these examples, it does not seem to provide any insight, while at the same time requiring additional assumptions that seem catered to the examples at hand.

stalemate assumption), but also with BC.

Suppose that, in a case of symmetry, two potential strengthened readings are expressible without added syntactic complexity. BC could then have the effect that *neither* implicature should be available. This is because each of the potential strengthenings could be blocked by the assertion of the competitor which encodes the same meaning without need for strengthening. We could call this configuration a case of *symmetric blocking*.

Swanson (2010) presents examples which he argues provide exactly such a scenario: Two stronger but mutually exclusive strengthened meanings of an expression have lexicalized counterparts. However, in contrast to the prediction of BC, Swanson observes that one of the two conceivable implicatures is possible in these cases. If Swanson's assumptions about the lexical meanings of the words involved are correct, this would falsify the blocking constraint. Let's consider one of the examples:

(70) Going to confession is permitted.

- (71) a. Going to confession is optional.
b. Going to confession is required.

Sentence (70) can implicate (71a), but not (71b). It can convey that confession is not required (and hence optional), but it cannot convey that that confession is not optional (and hence required). Swanson uses this and related examples to show that CF cannot be the only constraint on Excl. In our terminology, *Going to confession is optional* poses a simplex threat for the account based on CF.

We add here the observation that the example is *also* a straightforward counterexample to the stalemate assumption. The inference in (71b) is possible although it has a symmetric counterpart. Since all alternatives are of equal complexity, a formal filter on alternative will not be able to break the stalemate here, further supporting our claim that the stalemate assumption is incorrect.

If the lexical meaning of *optional* really lexically encodes *permitted and not required*, then (70) also falsifies BC, however: BC correctly predicts that the assertion with *require* blocks the unavailable strengthening of *permit* to convey 'required;' but the potential assertion with *optional* should block the available strengthening that confession is *not* required. Much like the stalemate assumption, BC would predict in this case that *both* conceivable implicatures are unavailable, contrary to fact.

We contend, however, that *optional* does not actually have this lexical meaning. If it did, both of the following statements should be equally infelicitous, but only the second is outright contradictory:⁸

- (72) a. ?Doing drugs is optional, and in fact it is not permitted
b. #Doing drugs is optional, and in fact it is required

This suggests that the lexical meaning of *optional* does not actually entail *permitted*. We submit that it conveys *permitted* in many (possibly most) contexts as an implicature. One informant called

⁸The following example found online supports this point:

- (i) Breaking the law is optional and the consequential punishment is known so what's the problem?

(72a) ‘cheeky,’ indicating that it is hard not to draw this implicature. But there is a clear contrast to (72b).⁹

Embedding these predicates under negation, where implicatures often vanish, provides further evidence for this conclusion. In the following examples, the preferred interpretation is the implausible reading that taking drugs is required—this is expected if the lexical meaning of *optional* is closer in meaning to *not required*:

- (73) a. Taking drugs is not optional.
 b. Taking drugs is non-optional

The relation between *optional* and *permitted* might therefore be parallel to the relation between *some* on the one hand and *not all* (and *some not*) on the other: The strengthened readings of *some* and *not all* are truth-conditionally equivalent (both convey *some but not all*, but neither basic meaning is equivalent to the strengthened reading of the other. Hence BC correctly predicts that each of them can be used to convey *some but not all*.¹⁰

If true, this diffuses the challenge for BC. However, these examples still contradict both CF and the stalemate assumption: Even if the basic meaning of *optional* does not entail *permitted*, the alternative poses a simplex threat for CF; and the very possibility of an implicature here appears to contradict the stalemate assumption.

Swanson (2017) discusses some other cases. Consider the following:

- (74) a. The heater sometimes squeaks.
 b. The heater intermittently squeaks.
 c. The heater occasionally squeaks.
 d. The heater always squeaks.

Why can *The heater sometimes squeaks* not implicate that the heater always squeaks, on the grounds that one or both of the alternatives based on *occasionally* and *intermittently* are denied?

It seems to us that *occasionally* may in fact be equivalent in meaning to *sometimes*. This would explain why (75a) sounds incoherent.

- (75) a. #The heater squeaked sometimes but not occasionally.
 b. ?The heater squeaked sometimes but not intermittently.

The example in (75b) with *intermittently*, on the other hand, seems comparatively coherent. Even though it’s a bit unclear what this sentence exactly conveys, it seems that the sentence can still implicate that the heater didn’t always squeak. Maybe *intermittently* implies a richer temporal

⁹Swanson (p.c.) observes that it is possible to use *optional* in contexts in which it is presupposed that something is not required, suggesting that it is not exactly equivalent to *not required*.

¹⁰Swanson also discusses the triplet *possibly, contingently, necessarily*. We think *contingently* might be similar to *optional*, in that rather than lexicalizing *possibly but not necessarily*, it might only encode *not necessarily*, and implicate *possibly*. Judgments are (even) murkier here, because *contingently* seems restricted to certain registers, and has a variety of more technical meanings. For example, the following sentence from an online discussion of Behaviorism uses *contingently* in a difference sense: *That is, what if a behavior is non-contingently reinforced? In this case reinforcers are presented to the subject at random.*

structuring of the squeakings that, even if denied, leaves room for the heater to not always be squeaking. This could explain why the implicature from *some* to *all* is not available here.¹¹

To summarize, we looked at several cases where BC seems to wrongly predict mutual blocking. After closer scrutiny of these examples, we are not convinced that these are in fact counterexamples to BC. However, even if our conclusions about the lexical meanings of the expressions are correct, they remain counterexamples to CF and the stalemate assumption.

We have not been able to find any sets of example that show ‘symmetric blocking,’ where two strengthenings based on symmetric alternatives are each lexicalized. There are two reasons that could explain this, even if BC is correct. The first is that we haven’t looked hard enough; the second is that such lexicalization patterns are not possible.

The case of *optional* is particularly interesting. If our conclusion about its lexical meaning is correct it shows that, contrary to what is sometimes assumed (starting with Horn 1972), expressions that play an equivalent role to *not all/some not* vis-à-vis their alternatives can be lexicalized (even if *optional* does not lexicalize the strengthened reading *permitted but not required*).

6 Conclusion

Symmetry arises when a statement *S* has a set of potential alternatives *Alt*, but excluding all of them would lead to a contradiction.¹² The symmetry problem is the question of why, in cases of symmetry, certain implicatures are sometimes attested while others are impossible.

The prevalent view is that symmetry generally leads to a stalemate, and that symmetry ties can only be broken in reference to the formal realization of the alternatives. We presented evidence that one particular instance of such a theory, the syntactic complexity filter (CF) (Katzir, 2007; Fox and Katzir, 2011), is problematic for two reasons. First, we showed that CF is not a general constraint on alternatives, as claimed, but at best applies only to sets of symmetric alternatives as a tie breaker. Second, CF is not general enough, and fails to rule out alternatives that are undesirable but syntactically simplex. We also argued against the underlying assumption that symmetry leads to stalemates, which plays a crucial role in making the CF viable in the first place.

We argued instead that the alternative meanings in *Excl* can, in principle, be freely selected. To account for why certain implicatures are nevertheless unavailable, we proposed several pragmatic constraints. Most important to the argument here was the blocking constraint BC, which constrains the *output* of strengthening. BC prohibits strengthenings if there is an utterance whose basic meaning encodes the strengthened meaning, and is at most as complex as the asserted utterance. This blocking constraint might itself be derivable from the Gricean maxim of manner. We also presented evidence showing that this constraint is not sufficient to explain all absent implicatures. Some implicatures appear to be unavailable simply because they are not contextually relevant; others because the implicature would not be contextually recoverable.

¹¹Whatever (75b) exactly conveys, it is clear that this implicature is not readily available for the sentence *the heater sometimes squeaks* without prior mention of *intermittently*. But that might be simply because the notion *intermittently* is not frequent enough to come to mind.

¹²Another, logically equivalent way of stating this is that symmetry arises if *S* entails the disjunction of its alternatives, see appendix B for a discussion of different notions of symmetry.

The blocking analysis does not necessitate the assumption that the excluded alternatives in Excl are syntactic expressions. While BC, too, is sensitive to the linguistic resources of a language, this sensitivity to form only comes in in the formulation of an output condition on strengthenings—no formal constraint on the set of alternatives that strengthening is based on is needed.

We hope to have shown that the alternative account in terms of blocking has the potential to provide an insightful account of the symmetry problem. The empirical ground we covered in this paper is still limited, and more work will be needed to test these ideas further.

A Blocking generalized: maximize basic meaning

Our blocking constraint BC stated in Section 1, repeated in (76) from (14), has the intended effect for the particular exclusion sets discussed in the main text. However, this rendition of constraint turns out to not apply correctly in the general case. We will show that it undergenerates blocking once a broader range of exclusion sets is considered, and propose an amended version that removes the undergeneration.

- (76) Blocking constraint (BC):
 $\text{Exh}_{\text{Excl}}(\phi)$ is blocked if: $\exists \psi [\text{Basic}(\psi) = \text{Exh}_{\text{Excl}}(\phi) \wedge \psi \text{ is no more complex than } \phi]$

To illustrate the undergeneration of blocking, consider once again sentence (1). (1), repeated here as (77a), can be used to exclude the proposition $\text{ALL}(\text{P})(\text{R})$, but not the proposition $\neg \text{ALL}(\text{P})(\text{R})$. We proposed that this is because of the existence of expression (77b), which is equivalent in meaning to the unavailable strengthened reading, and no more complex than (77a).

- (77) a. Some players responded.
 b. All players responded.

Uncontroversially, though, (77a) *can* be understood as excluding relevant propositions other than $\text{ALL}(\text{P})(\text{R})$. For example, it seems easy to imagine a context where *Some players responded* would be taken to also imply that the coach did not respond, $\neg \text{R}(\text{c})$. This leads us to conclude that $\text{R}(\text{c})$ is a potential member of the exclusion set. In turn, this leads one to consider the potential strengthening of (77a) in (78a), where $\text{R}(\text{c})$ is excluded in addition to $\neg \text{ALL}(\text{P})(\text{R})$.

- (78) a. $\text{Exh}_{\{\neg \text{ALL}(\text{P})(\text{R}), \text{R}(\text{c})\}}(1) = \text{ALL}(\text{P})(\text{R}) \wedge \neg \text{R}(\text{c})$
 b. $\text{Basic}((77\text{b})) \neq \text{ALL}(\text{P})(\text{R}) \wedge \neg \text{R}(\text{c})$

The crucial observation is now that, as stated in (78b), the potential strengthened meaning of (77a) in (78a) is *not* equivalent to the basic meaning of *All players responded*. Moreover, while $\text{ALL}(\text{P})(\text{R}) \wedge \neg \text{R}(\text{c})$ is the basic meaning of, say, sentence (79), this sentence is more complex than (77a), and the same is arguably true for any sentence with the same basic meaning. As a consequence, BC as stated allows for the strengthening resulting a meaning paraphrased in (79).

- (79) All players responded, but the coach did not.

The intuition that (77a) cannot imply $\text{ALL}(\text{P})(\text{R})$ is surely independent of what else the sentence may or may not be taken to imply beyond its basic meaning. In particular, the implication $\text{ALL}(\text{P})(\text{R})$ is not intuited to become available for (1) on the mere grounds that the sentence is *also*

understood to imply $\neg R(c)$. We therefore conclude that BC as formulated in (76) is too permissive, as the intended prohibition applies in too narrow a set of circumstances.

However, we propose that the blocking approach permits a natural generalization that removes this shortcoming. Note first that, as recorded in (80), the problematic meaning $ALL(P)(R) \wedge \neg R(c)$ could well be the strengthened meaning of the sentence *All players responded*, obtained by exclusion of just $R(c)$. So the problematic meaning could have been expressed by a sentence with a stronger basic meaning than (77a). We propose that it is on those grounds that it cannot be expressed by (77a) itself.

$$(80) \quad \text{Exh}_{\{R(c)\}}(77b) = ALL(P)(R) \wedge \neg R(c)$$

So we propose that blocking of a given strengthening is not just supported by sentences whose basic meaning is equivalent to the strengthened meaning. Instead, we hold that the correct blocking condition should ensure more broadly that the contribution of basic meaning to the overall utterance meaning is maximized, so that the contribution added by strengthening is minimized. The revised rendition of BC in (81) captures this idea.

(81) Blocking constraint (BC), revised:

$$\text{Exh}_{\text{Excl}}(\phi) \text{ is blocked if: } \exists \psi [\text{Exh}_{\text{Excl}}(\phi) \subseteq \text{Basic}(\psi) \subset \text{Basic}(\phi) \wedge \psi \text{ is no more complex than } \phi]$$

According to (81), a sentence ψ can block a strengthening of ϕ in virtue of having a basic meaning that is merely entailed by the strengthened meaning of ϕ , as long as the basic meaning of ψ is stronger than the basic meaning of ϕ , so that using ψ instead of ϕ will increase the contribution of basic meaning to the overall meaning. So generalized, BC has the effect that (77b) will correctly block the strengthening to $ALL(P)(R) \wedge \neg R(c)$ for (77a), on the grounds that the basic meaning of (77b), $ALL(P)(R)$, is entailed by this strengthened meaning, and at the same time is stronger than the basic meaning of (77a), $SOME(P)(R)$.

The proposed generalization of BC has a notable consequence for cases where two symmetric potential alternatives are compatible with one another. Consider, for example, the disjunctive sentence in (82), with the basic meaning in (83).

(82) Ann or Ben responded.

(83) Basic meaning of (82):

$$R(a) \vee R(b)$$

Beyond its basic meaning, sentence (82) could well be taken to imply that Ann and Ben did not both respond, $\neg(R(a) \wedge R(b))$, or that, say, the coach didn't respond, $\neg R(c)$. This indicates that both $R(a) \wedge R(b)$ and $R(c)$ can in principle be excluded in a strengthened meaning of (82). However, as recorded in (84), it is clear that (82) *cannot* be understood as strengthened via the exclusion of either of the disjuncts in (83), $R(a)$ or $R(b)$, to yield the strengthened meaning that Ann responded and Ben didn't or that Ben responded and Ann didn't.

(84) Unavailable strengthened meanings of (82):

- a. $(R(a) \vee R(b)) \wedge \neg R(a) = R(b) \wedge \neg R(a)$
- b. $(R(a) \vee R(b)) \wedge \neg R(b) = R(a) \wedge \neg R(b)$

The revised blocking condition in (81) predicts this limitation in available strengthenings. Note first that R(a) and R(b) are the basic meanings of sentences in (85a) and (85b), respectively.

- (85) a. Ann responded.
 b. Ben responded

Now, it holds that Basic(85a) is stronger than Basic(82) and at the same time is entailed by the potential strengthened meaning of sentence (82) in (84b). We moreover note that (85a) is no more complex, in fact less complex, than sentence (82). According to the generalized version of BC in (81), these observations entail that (85a) blocks the strengthening in (84b). For parallel reasons, BC likewise ensures that (85b) blocks the potential strengthening in (84a).

We note that the original blocking condition in (76) would arguably not prohibit the strengthenings in (84). The reason is that the basic meanings of (85a,b) are weaker than (84a,b), and it appears that any sentences whose basic meanings are equivalent to (84a,b) are more complex than (82).

In sum, our generalized BC can derive blocking in cases where the original version of BC has no effect, on the grounds that the exclusion of one of two symmetric alternatives does not yield a strengthened meaning that is equivalent to the basic meaning of the other. One possible reason for this non-equivalence is the presence of an additional member of the exclusion set, as in our initial illustration with example (77a). Another possible reason, illustrated with (82), is that the symmetric alternatives are not incompatible with one another. In either case, the exclusion of one of the alternatives yields a meaning that is not entailed by the basic meaning of the other.

This approach to disjunctive sentences is aligned with a proposal in Stalnaker (1975), a precedent for the idea that these strengthenings of disjunctions are unavailable due to blocking: “...the disjunction would be equivalent to the assertion of one of the disjuncts alone. So the disjunctive assertion would be pointless, hence misleading, and therefore inappropriate.” More broadly, like the initial version of BC, the generalized rendition is aligned with the approach in Rational Speech Act Theory (RSA), briefly referenced in Section 1 (e.g. Goodman and Frank 2016, Degen 2023). On that view, maximization of basic meaning in the overall meaning conveyed can be understood as maximization of the likelihood that the intended meaning is communicated successfully.

At the same time, for the case of disjunctions, our generalized BC in (81) is in opposition to proposals that directly posit constraints on the exclusion set. In particular, the unavailability of the strengthenings in (84) for the disjunction in (82) could be viewed as a consequence of the stalemate condition discussed in Section 4 (cf. Katzir 2007, Fox and Katzir 2011). By banning exclusion sets that contain just one of R(a) and R(b), the stalemate condition would correctly predict that symmetry cannot be broken in this case. However, given the challenges to the stalemate assumption presented in Section 4, we suggest that its potential benefits for disjunctions are actually better understood in terms of blocking.¹³

¹³The literature on implicature has also appealed to blocking in order to ensure that a less complex expression will be chosen over a more complex expression with an equivalent basic meaning, even in the absence of strengthening. McCawley (1978) and Horn (1989) discuss a preference for more prolix expression in the context of scalar implicatures, relating this effect to the Paninian “Elsewhere Principle” (Kiparsky, 1973). See Mayr and Romoli (2016) and Meyer (2016) for recent discussions of related blocking conditions that favor more prolix expressions. We will not discuss such cases here.

B On the logical relation between CF and BC under symmetry

CF and BC, both repeated below from Section 1, are very different conditions, as is evident from the fact that only BC appeals to how an asserted sentence relates to alternative sentences in terms of meaning.

(86) Complexity filter on Excl (CF):

$$\text{Excl}(\phi) \subseteq \{\text{Basic}(\psi): \psi \text{ is no more complex than } \phi\}$$

(87) Blocking constraint (BC):

$$\text{Exh}_{\text{Excl}}(\phi) \text{ is blocked if: } \exists \psi [\text{Basic}(\psi) = \text{Exh}_{\text{Excl}}(\phi) \wedge \psi \text{ is no more complex than } \phi]$$

At the same time, though, both CF and BC somehow reference alternatives' structural complexity. For cases where alternatives are symmetric, this has the effect that CF and BC can under certain conditions coincide in their predicted effects on possible strengthenings. Below, we will briefly discuss those conditions, and also how predictions of CF and BC differ when the conditions are not met.

We have characterized alternatives as *symmetric* relative to a sentence ϕ if the strengthening of $\text{Basic}(\phi)$ obtained by excluding of each of those alternatives would yield a contradiction. With symmetry so understood, alternatives are symmetric relative to ϕ just in case their disjunction is entailed by $\text{Basic}(\phi)$. Confining attention to cases with just two alternatives, $a1$ and $a2$, the assumption that those are symmetric relative to ϕ can then be stated as in (88).

$$(88) \quad \text{Basic}(\phi) \subseteq a1 \vee a2$$

It will actually be instructive for our purposes to consider the two further assumptions in (89). (89i) states that the disjunction of $a1$ and $a2$ entails $\text{Basic}(\phi)$, which together with (88) yields equivalence of this disjunction with $\text{Basic}(\phi)$. (89ii) states that $a1$ and $a2$ are contraries, that is, are incompatible with each other, so that given (88), the meaning that results from excluding either of them is guaranteed to entail the other.

$$(89) \quad \begin{array}{ll} \text{i.} & a1 \vee a2 \subseteq \text{Basic}(\phi) \\ \text{ii.} & a1 \wedge a2 = \emptyset \end{array}$$

Notably, the conjunction of the statements in (88) and (89i,ii) is equivalent to the assumption that relative to $\text{Basic}(\phi)$, $a1$ and $a2$ are complementary to each other. That is, it follows from (88) and (89i,ii) that $a1$ and $a2$ relate to $\text{Basic}(\phi)$ as stated in (90i,ii).

$$(90) \quad \begin{array}{ll} \text{i.} & \text{Basic}(\phi) \wedge \neg a1 = a2 \\ \text{ii.} & \text{Basic}(\phi) \wedge \neg a2 = a1 \end{array}$$

It will be useful for us to introduce terminology to facilitate reference to the assumptions in (89). If $a1$ and $a2$ are symmetric relative to ϕ , meeting the condition in (88), and in addition relate to ϕ as in (89i), we will say that $a1$ and $a2$ are *entailment symmetric* relative to ϕ . If (89ii) holds in addition to (88), we will say that $a1$ and $a2$ are *contrary symmetric* relative to ϕ . And if $a1$ and $a2$ are both entailment symmetric and contrary symmetric relative to ϕ , we will also say that they are *strictly symmetric* relative to ϕ . Note that this terminology differs from the one introduced by Fox and Katzir (2011), who use bare *symmetry* to refer to the relation of strict symmetry.

We can now show, that in cases of strict symmetry, the symmetry breaking effects of CF and BC coincide. That is, CF breaks symmetry in a given direction just case case BC does, so if one of the constraints CF and BC permits (91i) but prohibits (91ii), the same will be true for the other constraint.

- (91) i. $\text{Basic}(\phi) \wedge \neg a1$
 ii. $\text{Basic}(\phi) \wedge \neg a2$

To facilitate our demonstration of this point, let us say that an alternative *meaning* is “simplex” if it permits a linguistic encoding no more complex than ϕ , and “complex” otherwise. Now, assuming strict symmetry of $a1$ and $a2$ relative to ϕ , suppose that CF permits the strengthened meaning (91i) but prohibits (91ii). Since CF permits (91i), we can conclude that $a1$ must be simplex, hence that that there is a sentence ψ no more complex than ϕ such that $\text{Basic}(\psi) = a1$. Given strict symmetry, in particular the equality in (90ii), this entails that BC precludes the strengthening (91ii). At the same time, since CF prohibits (91ii), we can conclude that $a2$ is complex. This ensures that BC cannot prevent the strengthening (91i), the equality in (90i) notwithstanding. In sum, we can conclude that like CF, BC permits (91i) and prohibits (91ii).

Conversely, still assuming strict symmetry, suppose now that BC permits (91i), but prohibits (91ii). Given strict symmetry, hence in particular the equality in (90ii), BC can prohibit (91ii) only if $a1$ is simplex, which in turn guarantees that CF permits (91i). At the same time, for BC to permit (91i), given the equality (90i), it must be that $a2$ is complex, which in turn ensures that CF prohibits (91ii). In sum, we can conclude that like BC, CF permits (91i) and prohibits (91ii).

So as long as a pair of alternatives is strictly symmetric relative to an asserted sentence, they are necessarily aligned in their effects on symmetry breaking. To illustrate this alignment with a concrete example, CF and BC alike predict correctly that (92) can be strengthened by excluding simplex $\text{Basic}(92a)$ to yield $\text{SOME}(P)(R) \wedge \neg \text{ALL}(P)(R)$. But assuming that the meaning $\text{Basic}(92b)$ is complex, as suggested by the complexity of sentence (92b) itself, both CF and BC predict correctly that strengthening to $\text{ALL}(P)(R)$ by excluding $\text{Basic}(92b)$ is not attested.

- (92) Some players responded.
 a. All players responded
 b. Some but not all players responded.

Even under the assumption of strict symmetry, however, the effects of CF and BC do not fully coincide. What we know is that if either of CF and BC fails to break symmetry, then so must the other. However, this failure of symmetry breaking manifests itself differently with CF and BC. Each constraint prohibits what the other permits, and vice versa. Here is why. If CF permits both (91i) and (91ii), then this entails that both $a1$ and $a2$ are simplex. In turn, given the equalities in (90), this means that BC will prohibit both (91i) and (91ii). Conversely, if BC permits both (91i) and (91ii), then given the equalities in (90), it follows that both $a1$ and $a2$ are complex, which in turn entails that CF will prohibit both (91i) and (91ii).

This sort of complementarity between CF and BC is potentially illustrated by cases like (93) repeated from Section 5. Under the assessment in Swanson (2010), $\text{Basic}(93a)$ and $\text{Basic}(93b)$ are complementary relative to (93), instantiating strict symmetry. Recall that under this assumption, given that both $\text{Basic}(93a)$ and $\text{Basic}(93b)$ are simplex, CF allows for either to be excluded,

whereas BC prohibits the exclusion of either. Of course, both potential predictions are equally unwanted, as symmetry is intuited to be broken by the exclusion of Basic(93a). Therefore, the difference in prediction in this case cannot help us choose between CF and BC. In fact, we are not aware of any cases with strictly symmetric alternatives that would adjudicate between CF and BC on empirical grounds.

- (93) Going to confession is permitted.
- a. Going to confession is required.
 - b. Going to confession is optional.

Let us now turn to looking into predictions that hold if symmetry is not assumed to be strict. To begin, suppose we merely conjoin (88) with the condition in (89i), so that symmetry is still entailment symmetric, but not necessarily contrary symmetric, hence not necessarily strict. To be sure, with CF and BC as stated in (86) and (87), the results established above under the assumption of strict symmetry cannot be expected carry over in full to that case. However, those results *do* carry over after all as soon as BC is generalized as shown in Appendix A, as long as the basic meanings of the alternatives are independent. We refrain from detailing a proof of this claim here. Instead, we content ourselves with illustrating just one of the parallels with strict symmetry, namely with the case in (70) under the assumptions in Swanson (2010). That is, we will illustrate that under the assumption of mere entailment symmetry, if no symmetry breaking is predicted, then BC prohibits what CF permits. This is illustrated by observations about the example in (94), repeated from Appendix A.

- (94) Ann or Ben responded.
- a. Ann responded.
 - b. Ben responded.

Since the disjunction of Basic(94a) and Basic(94b) is not merely entailed by Basic(94), but equivalent to it, this case instantiates both (88) and (89i), hence instantiates entailment symmetry. Now, given that the alternatives (94a) and (94b) are both simplex, CF by itself permits the exclusion of either Basic(94a) or Basic(94b). In contrast, as established in Appendix A, the generalized version of BC prohibits strengthening of (94) by excluding either Basic(94a) or Basic(94b). Notably, as noted in Appendix A, given that intuitively neither alternative can be excluded, this is a potential advantage for BC.

To conclude our survey of the symmetry typology, suppose now that we merely conjoin (88) with the condition in (89ii), so that symmetry is now assumed to be contrary, but not necessarily strict. It is in this case that CF and BC drift further apart in terms of their predictions—and yet at the same time continue to overlap, giving rise to a complex pattern. The overlap is illustrated by our example initially presented in Section 1 to introduce BC as an alternative to CF, repeated in (95).

- (95) Some players responded.
- a. All players responded.
 - b. Not all players responded.

Note that the alternatives in this case are not strictly symmetric, on the grounds that one of them, viz. Basic(95b) does not entail Basic(95). Regardless, as established in Section 1, CF and BC agree in their predictions about this case. Both correctly permit the exclusion of Basic(95a) to yield

$\text{SOME}(\text{P})(\text{R}) \wedge \neg \text{ALL}(\text{P})(\text{R})$, and both correctly prevent the exclusion of Basic(95b) that would yield $\text{ALL}(\text{P})(\text{R})$. Close inspection reveals that the alignment of CF and BC in this case is crucially enabled by the fact that despite the non-strict symmetry, one alternative, viz. Basic(95a), still entails Basic(95). For $\text{Basic}(95a) = a1$, this crucially supports the equality in (90ii), hence in a sense supports one side of strictness. It is in virtue of this one-sided strictness, in conjunction with the fact that the entailing alternative Basic(95a) happens to be simplex, that BC has the welcome effect of prohibiting the exclusion of Basic(95b).

Despite this overlap in predictions, however, entailment symmetry is insufficient to ensure that CF and BC agree in general in their predictions about symmetry breaking. This point is already established by the simplex threat data discussed in Section 3, such as the antonym example repeated in (96). As discussed in Section 3, given that both Basic(96a) and Basic(96b) are simplex, CF cannot prevent the exclusion of either, hence cannot break symmetry. In contrast, BC correctly prohibits the exclusion of (96b) while allowing for the exclusion of (96a).

- (96) Some players are inside.
- a. All players are inside.
 - b. Some players are outside.

Notably, where symmetry is merely contrary but not strict, predictions about symmetry breaking under CF and BC could also come apart in cases where one alternative is complex and the other simplex. Imagine, for example, a language in which *SOME* and $\neg \text{ALL}$ are realized with expressions of equal complexity, say *some* and *nall*, but the expression realizing *ALL* is more complex, say *not nall*. Suppose that the language does not differ from English in other relevant ways. CF and BC would then make different predictions about the example in (97).

- (97) Some players responded.
- a. Not nall players responded.
 - b. Nall players responded.

Given that Basic(97a) is complex while Basic(97b) is simplex, CF would allow for (97) to be strengthened by excluding (95b) to yield $\text{ALL}(\text{P})(\text{R})$, but would prohibit the exclusion of (95b) to yield $\text{SOME}(\text{P})(\text{R}) \wedge \neg \text{ALL}(\text{P})(\text{R})$. In contrast, BC would not break symmetry in this case, in virtue of permitting either direction of strengthening. Comparison with our discussion of the case in (95) reveals why this would be so: even though like (95), (97) involves an alternative that entails the assertion, viz. Basic(97b), this alternative is complex, rather than simplex as in the case of (95).

It bears note that we have not found any set of lexicalized meanings in English that would fit the hypothetical configuration in (97). There may be principled reasons for this. Horn (1972), for example, argued that negated universal meanings like $\neg \text{ALL}$ are never lexicalized. Given our discussion in Section 5, Swanson's (2010) case in (93) above poses a potential challenge to this conjecture. In terms of meaning, we think that simplex *optional* may well have the basic meaning $\neg \square$, and accordingly relate to its alternatives in the same way as complex *not all* does. But even if this is so, the second ingredient of our hypothetical configuration in (97) is missing. The reason is that $\square$, the counterpart of the basic meaning *ALL* of our hypothetical complex *not nall*, does not require a complex encoding, being the basic meaning of, say, *required*.

Taking stock, under a notion of symmetry that renders the disjunction of two symmetric alter-

natives equivalent to the assertion, CF and BC necessarily agree in the direction of symmetry breaking, although they may diverge in the specifics of their predictions about cases where they both predict that symmetry is not broken. With symmetry construed as more permissive, allowing for alternatives that do not entail the assertion, there is no simple link between CF and BC in terms of their effects on exhaustification, although predictions can still happen to coincide. In sum, under the mere assumption that alternatives are symmetric in the sense of the assertion entailing their disjunction, the effects of CF and BC can agree or diverge in complex ways. Relating the predicted pattern to the linguistic evidence, we also saw that some of the discriminating predictions are hard or impossible to evaluate on empirical grounds.

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